

Why the f*ck do we need amplifiers?

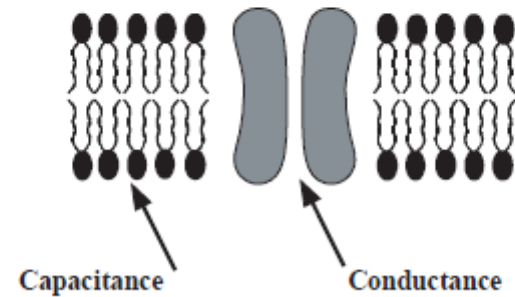
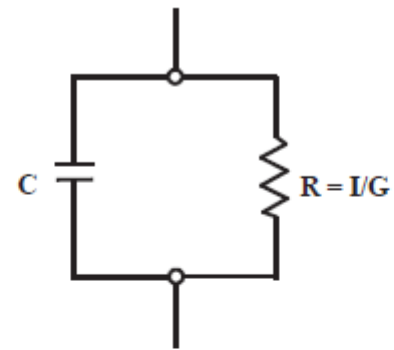


Figure 1.13: Membrane behavior compared with an electrical current.

The plasma membrane is a RC circuit

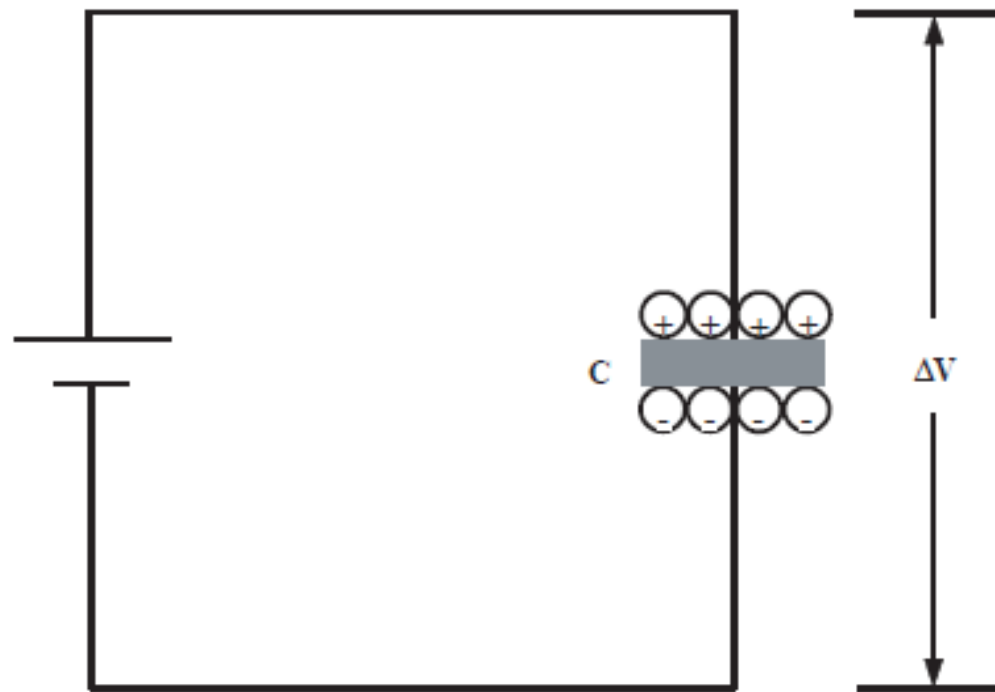
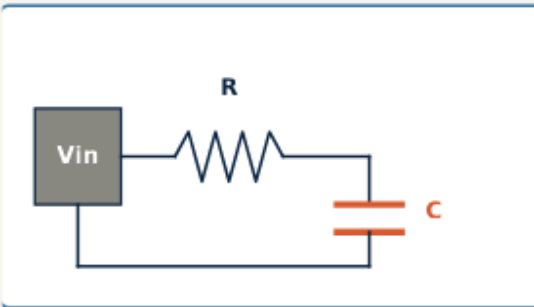


Figure 1.11: Capacitance. A charge Q is stored in a capacitor of value C held at a potential ΔV .



Charging ($V_{in} = V_0$)

$$V_C(t) = V_0(1 - e^{-t/RC})$$

$$I(t) = \frac{V_0}{R} e^{-t/RC}$$

$$Q(t) = C V_C(t)$$

Discharging ($V_{in} = 0$)

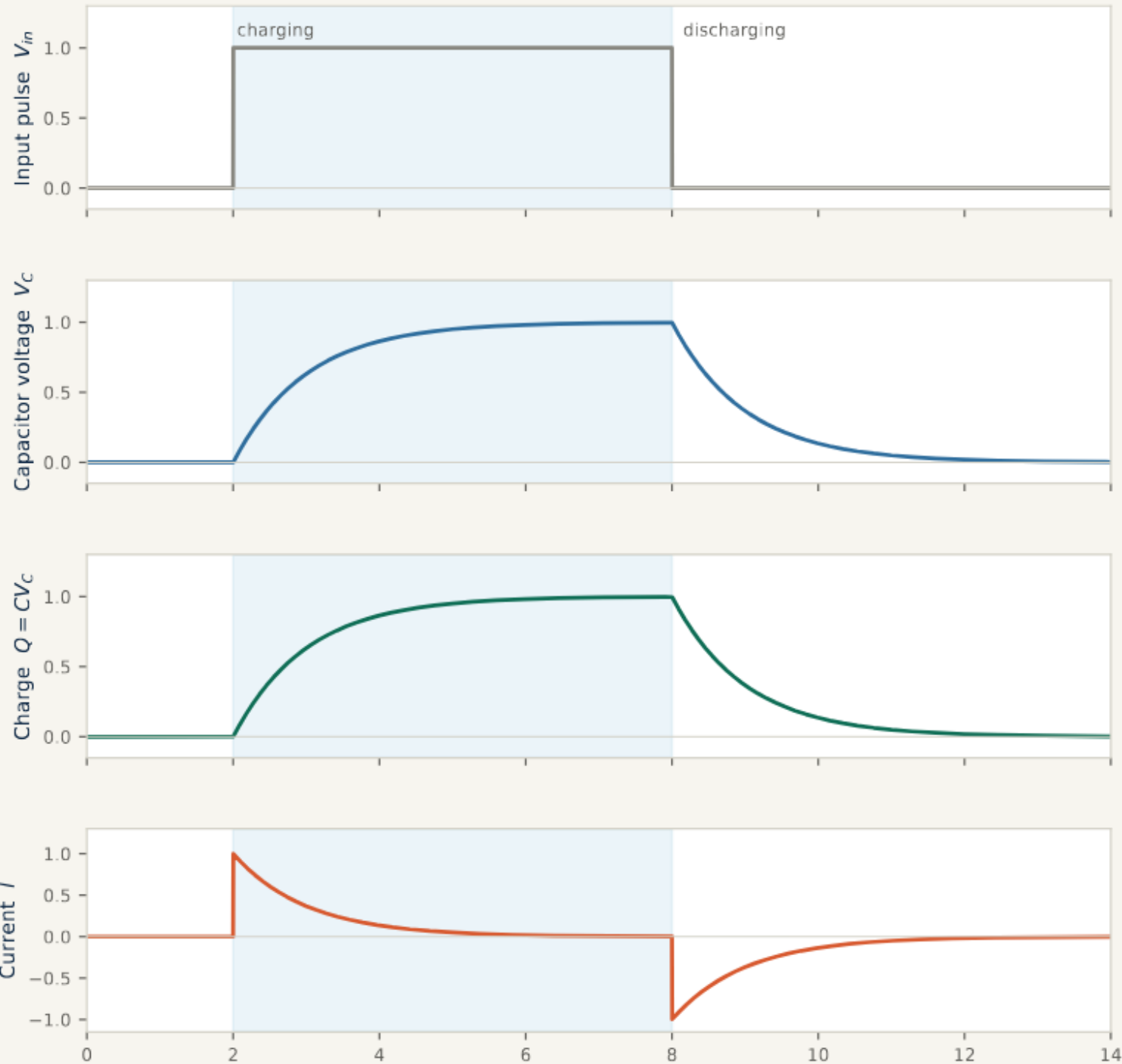
$$V_C(t) = V_0 e^{-t/\tau}$$

$$I(t) = -\frac{V_0}{R} e^{-t/\tau}$$

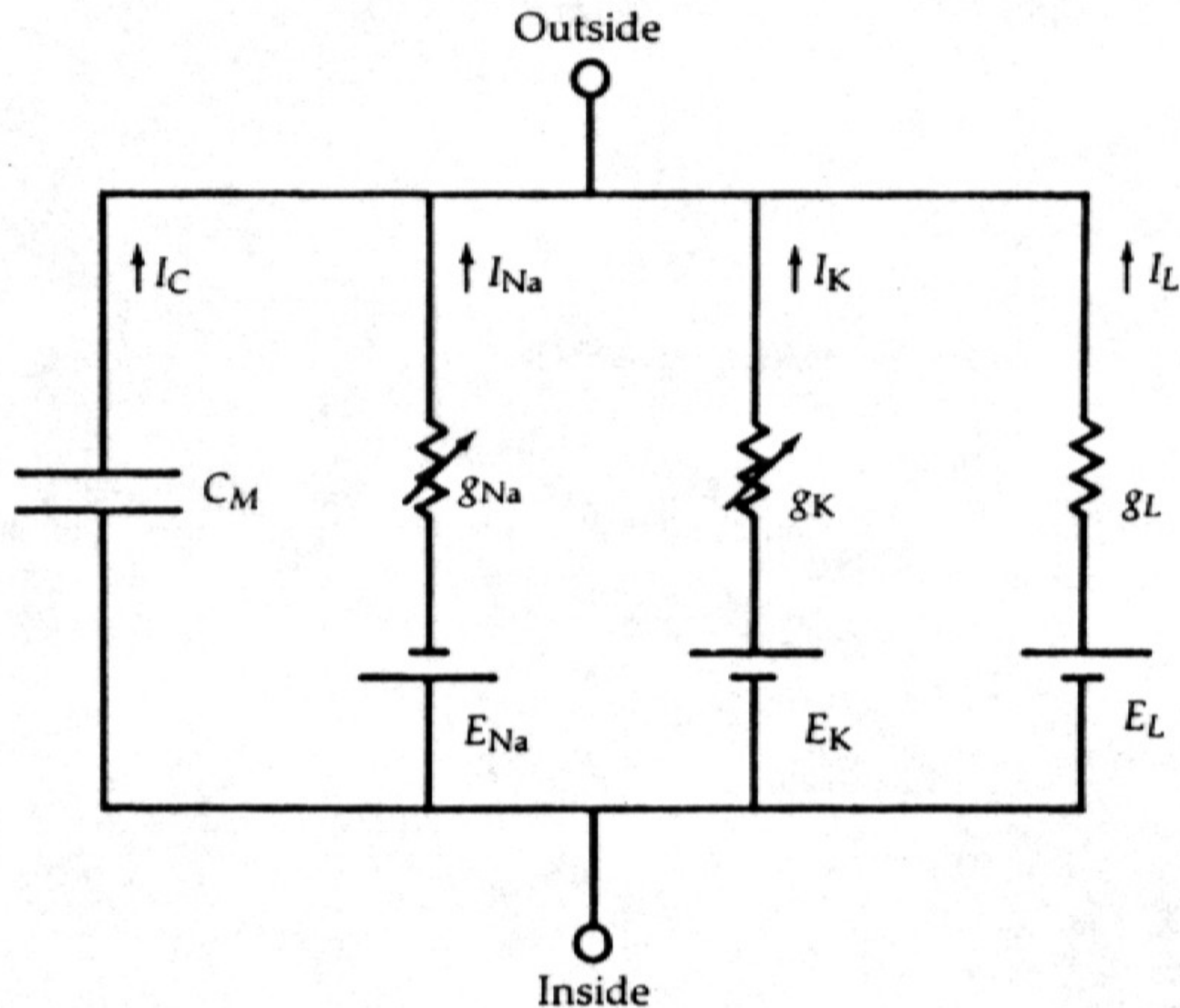
$$\tau = RC$$

Time to reach ~63% of the change (charging) or fall to ~37% (discharging).

Shown: $R = C = 1 \rightarrow \tau = 1$



Hodgkin and Huxley equivalent circuit



Why not use oscilloscopes?

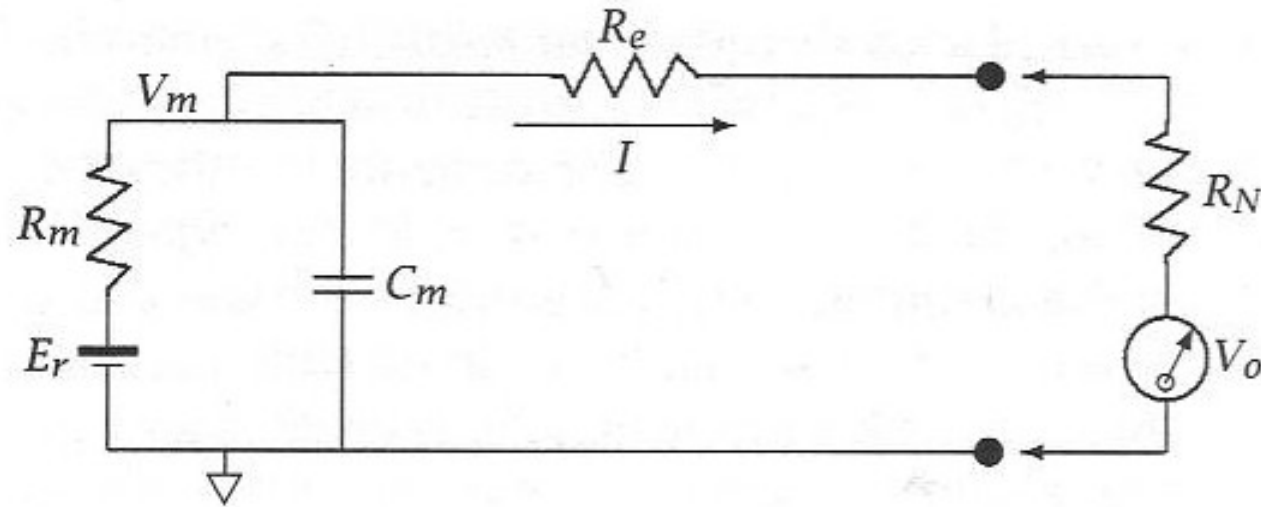


Figure A.23 Measuring the membrane potential (V_m) with a low input resistance device.

With representative values

$$V_o = \frac{R_N V_m}{R_e + R_N} = \frac{10^6}{1.01 \times 10^8} V_m$$

V_m attenuates 100 times !!!

Measuring biological signals

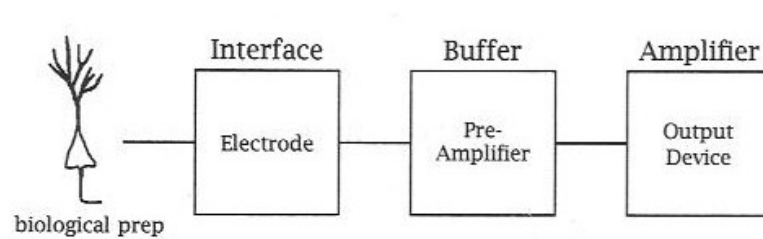
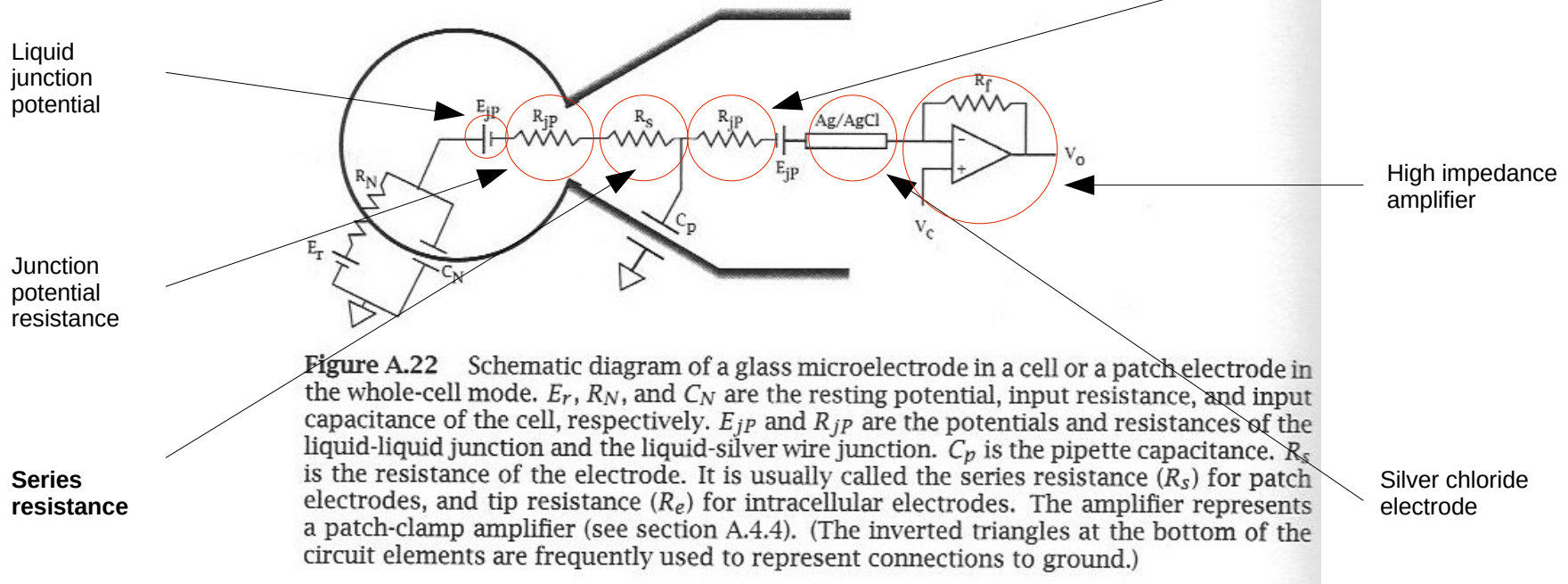
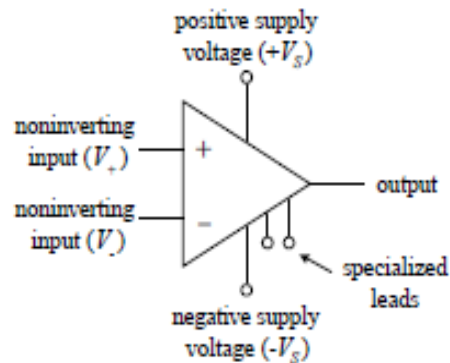


Figure A.21 Stages of a measuring system for biological signals.



La Resistencia en serie nos va a penar el resto de nuestra vida

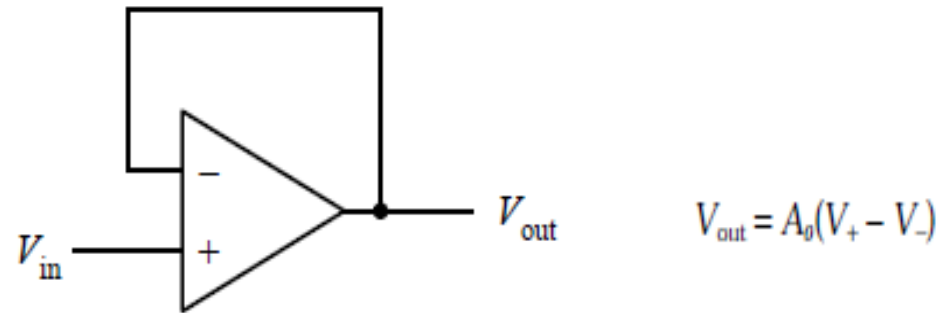
Operational Amplifier (OPA)



$$V_{\text{out}} = A_0(V_+ - V_-)$$

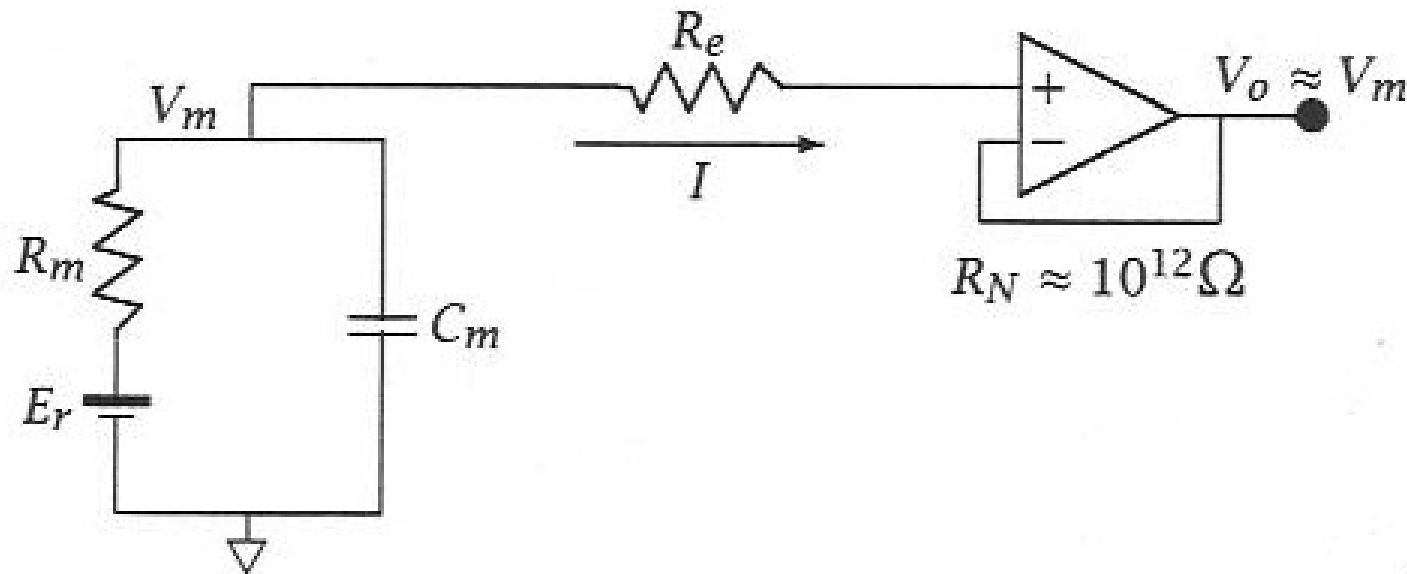
- Para un amplificador ideal $A = \infty$, para uno real $10^5 - 10^6$
- Para un amplificador ideal la impedancia de entrada es infinita, $R_{\text{in}} = \infty$. Para un amplificador real R_{in} es $10^8 - 10^{12}$ Ohm.
- Para un amplificador ideal la impedancia de salida es cero, $R_{\text{out}} = 0$, para uno real es 10-1000 Ohm.
- Las entradas de un amplificador ideal **no toman corriente**, las de uno real si lo hacen y se llama “input bias current”

Voltage follower (feedback loop)



- Cuando el OPA sensa una diferencia de potencial entre V_- y V_+ , este envia la corriente necesaria para mantener esta diferencia en zero

High input resistance



Voltage follower

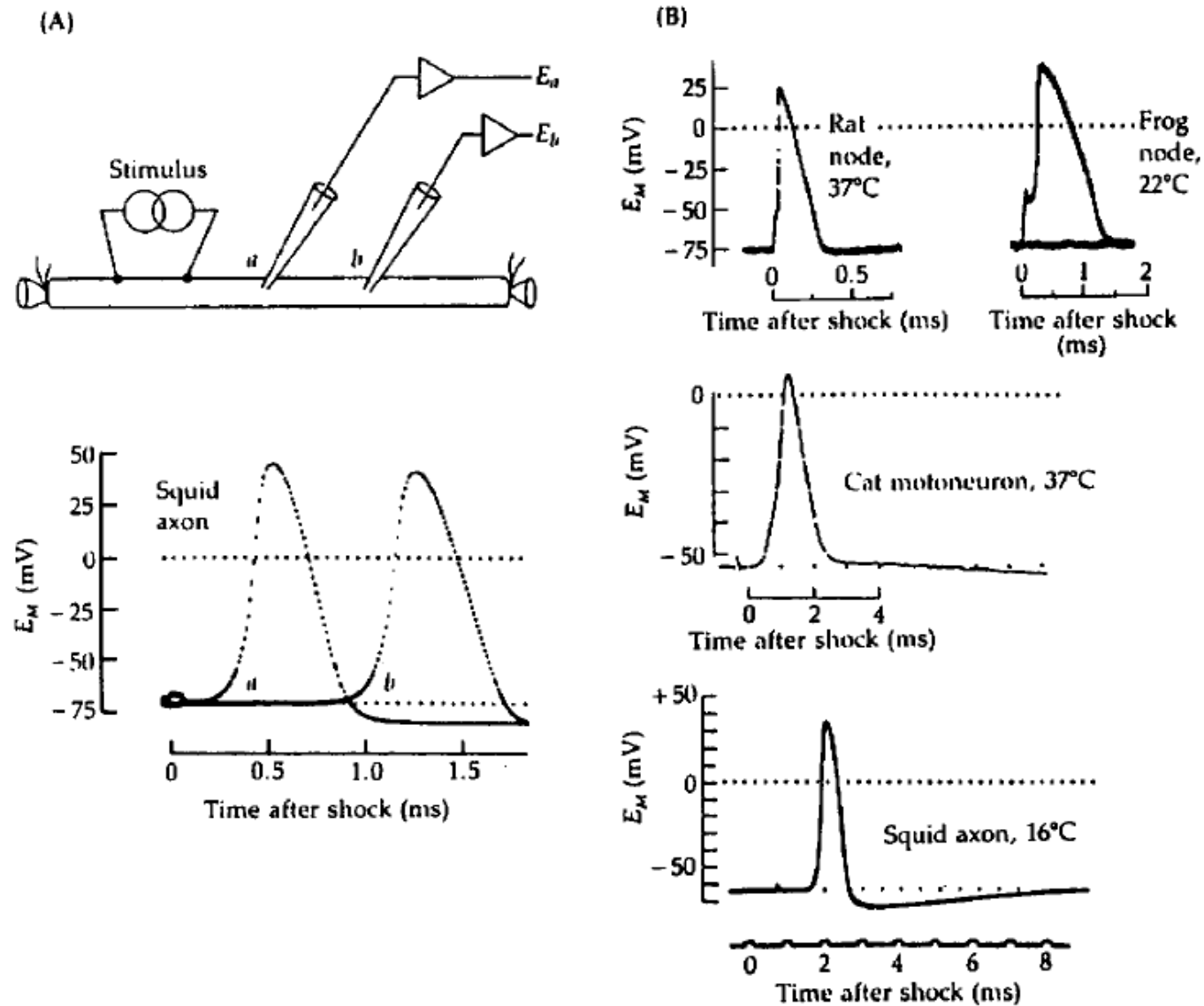
Figure A.24 Measuring V_m with high-impedance buffer amplifier.

With representative values

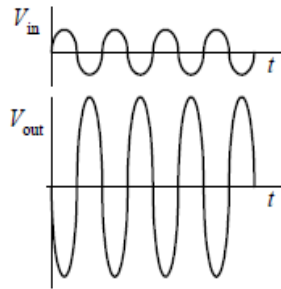
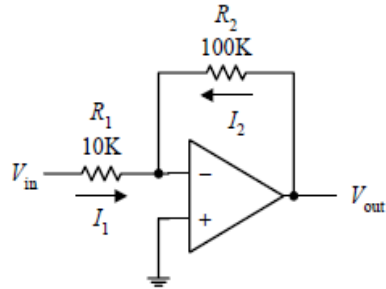
$$V_o = \frac{R_N V_m}{R_e + R_N} = \frac{10^{12}}{1.0001 \times 10^{12}} V_m \approx V_m.$$

Now we can measure the resting potential !!!!!

Measuring V_m with high input R amplifiers



Amplificador inversor



$$I_1 + I_2 = 0$$

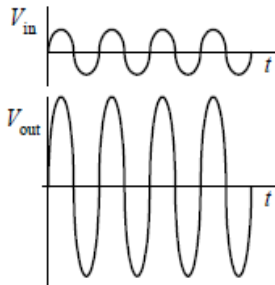
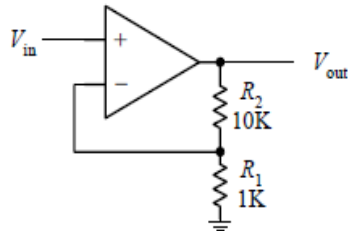
$$\frac{V_i - V_-}{R_1} + \frac{V_o - V_-}{R_2} = 0$$

$$V_o = -\frac{R_2}{R_1} V_i$$

$$\frac{V_i}{R_1} + \frac{V_o}{R_2} = 0$$

$$V_i - V_o$$

Amplificador no inversor



$$I_1 + I_2 = 0$$

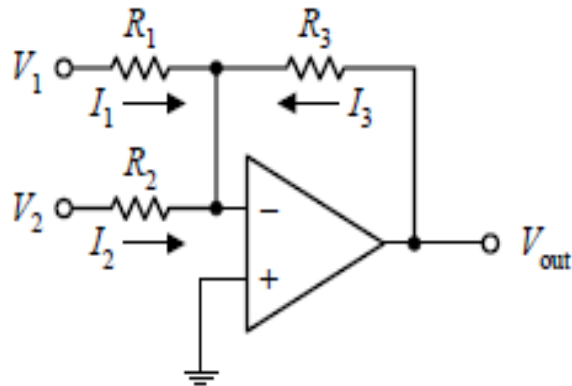
$$\frac{0 - V_-}{R_1} + \frac{-V_o + V_-}{R_2} = 0$$

$$V_o = \left(\frac{R_2}{R_1} + 1 \right) V_i$$

$$V_i = V_-$$

$$\frac{-V_i}{R_1} + \frac{V_o - V_i}{R_2} = 0$$

Amplificador sumador



$$I_1 = \frac{V_1 - V_-}{R_1} = \frac{V_1 - 0V}{R_1} = \frac{V_1}{R_1}$$

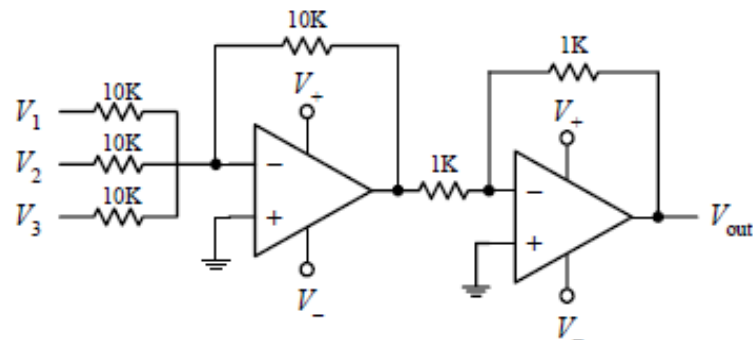
$$I_2 = \frac{V_2 - V_-}{R_2} = \frac{V_2 - 0V}{R_2} = \frac{V_2}{R_2}$$

$$I_3 = \frac{V_{out} - V_-}{R_3} = \frac{V_{out} - 0V}{R_3} = \frac{V_{out}}{R_3}$$

$$\frac{V_o}{R_3} + \frac{V_1}{R_1} + \frac{V_2}{R_2} = 0$$

$$V_o = -\left(\frac{R_3}{R_1}V_1 + \frac{R_3}{R_2}V_2\right)$$

Amplificador sumador + inversor

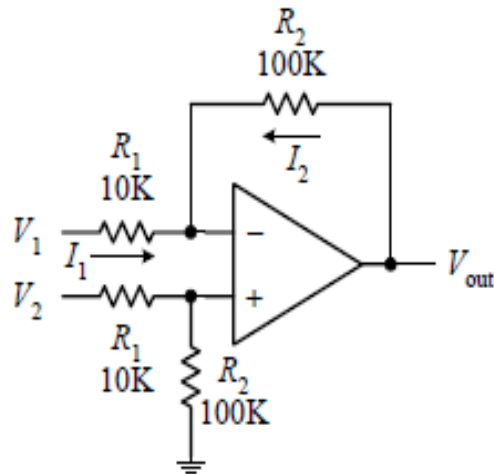


$$si R_1 = R_2 = R_3$$

$$V_o = -(V_1 + V_2)$$

DIFFERENCE AMPLIFIER

Determine V_{out} .



First, you determine the voltage at the noninverting input by using the voltage divider relation (again, assume that no current enters the inputs):

$$V_+ = \frac{R_2}{R_1 + R_2} V_2$$

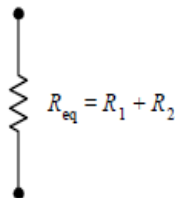
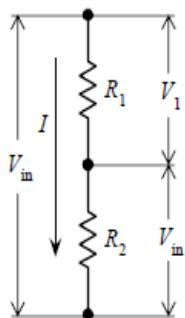
Next, apply Kirchhoff's current junction law to the inverting input ($I_1 = I_2$):

$$\frac{V_1 - V_-}{R_1} = \frac{V_- - V_{out}}{R_2}$$

Using rule 4 ($V_+ = V_-$), substitute the V_+ term in for V_- in the last equation to get

$$V_{out} = \frac{R_2}{R_1} (V_2 - V_1)$$

If you set $R_1 = R_2$, then $V_{out} = V_2 - V_1$.



$$I = \frac{V_{in}}{R_{eq}}$$

$$V_1 = \frac{R_1}{R_1 + R_2} V_{in}$$

$$V_2 = \frac{R_2}{R_1 + R_2} V_{in}$$

Voltage clamp

THE IDEAL VOLTAGE CLAMP

The ideal voltage clamp simply consists of a battery, a switch, a wire, the cell and an ammeter (Figure 3.10). Since the wire has zero resistance when the switch is closed, the membrane potential steps instantly to the battery voltage. This generates an impulse of current that charges the membrane capacitance, followed by a steady-state current to sustain the voltage across the membrane resistance.

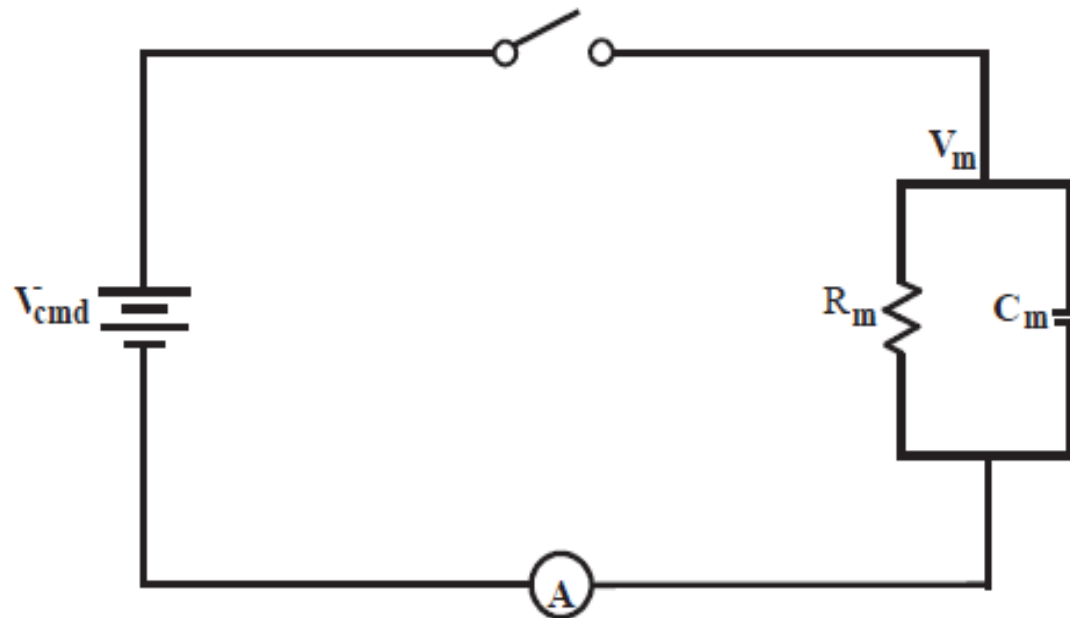


Figure 3.10: The ideal voltage clamp.

The ideal voltage clamp (Figure 1) consists of a voltage source providing the clamp potential V_C , the model membrane (membrane resistance R_M and capacitance C_M in parallel), a switch and an ampere meter for measuring membrane current I_M .

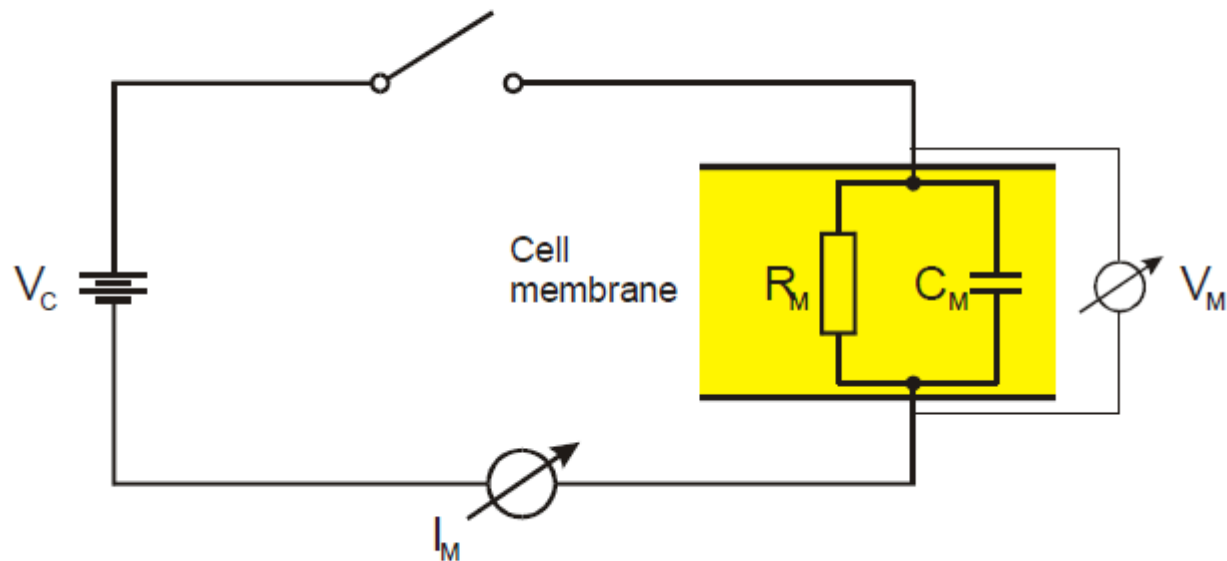


Figure 1: The ideal Voltage Clamp

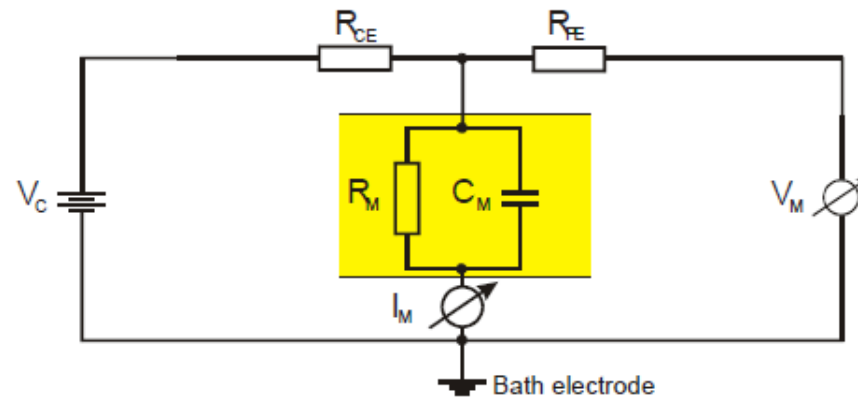


Figure 3: **Voltage-Clamp circuit with a current electrode (CE) and a voltage electrode (VE).**

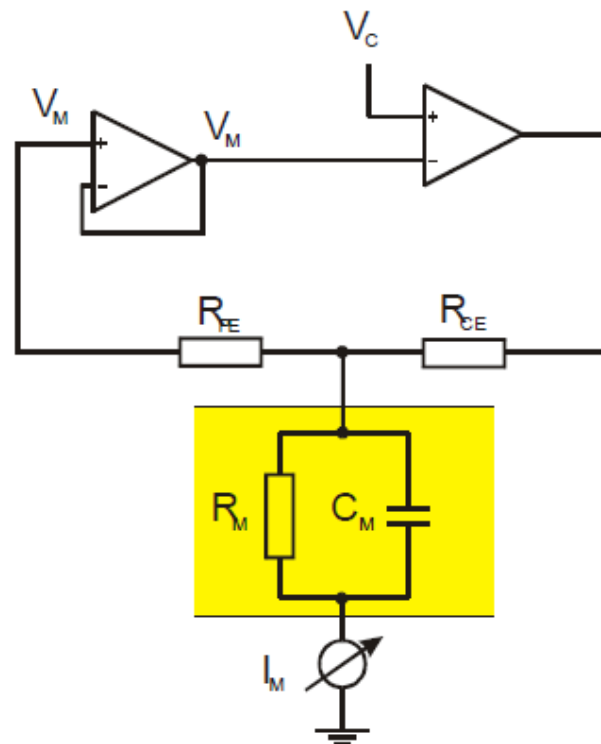
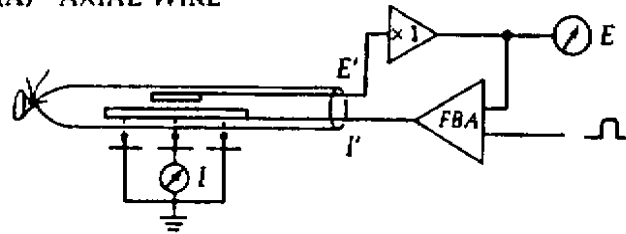


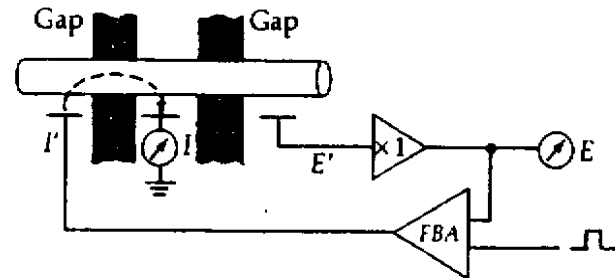
Figure 5: **Two-electrode voltage-clamp circuit using op-amps for measuring membrane potential and performing voltage clamp by a negative feedback circuit**

Voltage clamp

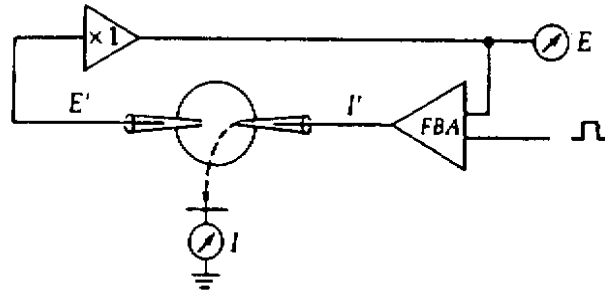
(A) AXIAL WIRE



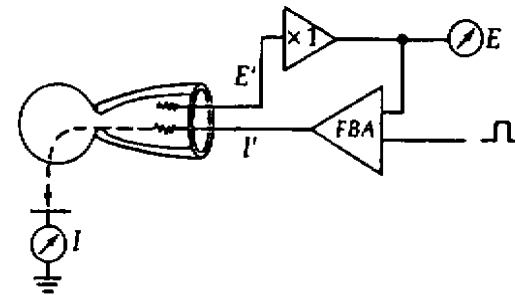
(B) DOUBLE GAP



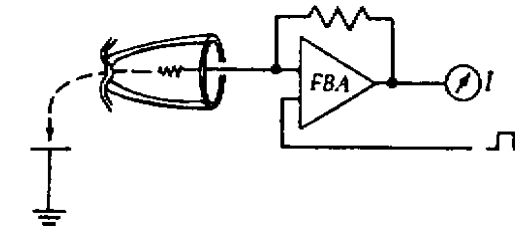
(C) TWO MICROELECTRODE



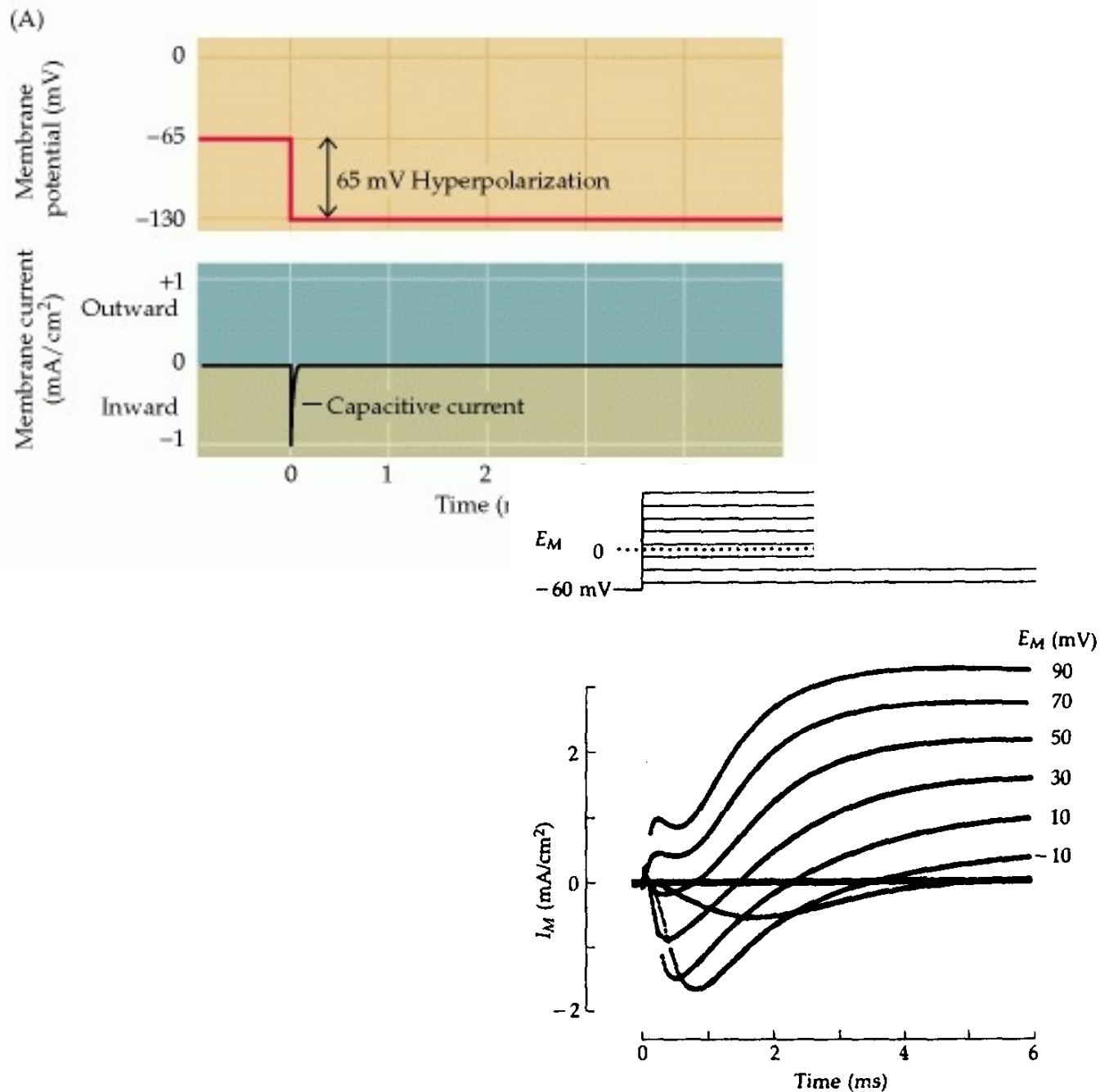
(D) SUCTION PIPETTE



(E) PATCH CLAMP



Voltage clamp en el axon gigante



Patch clamp

A.4.4 Patch-clamp circuit (current-to-voltage converter)

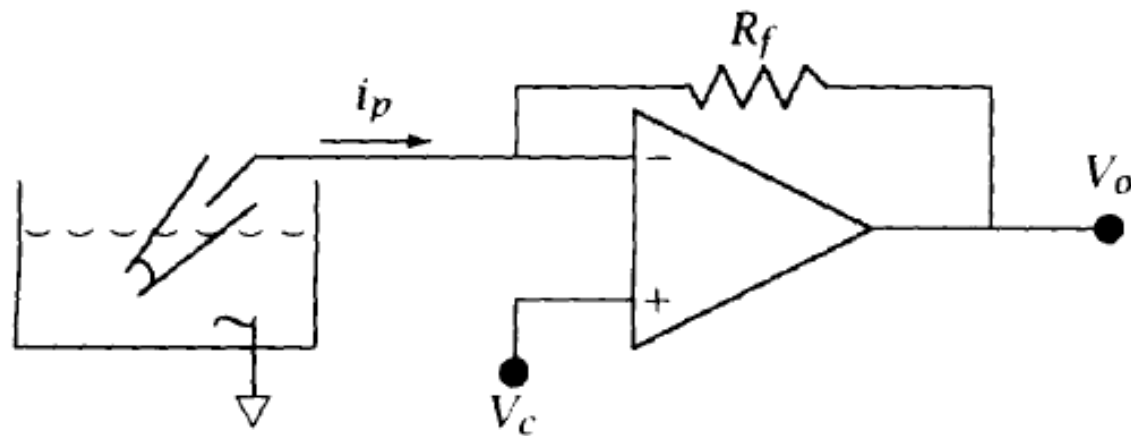
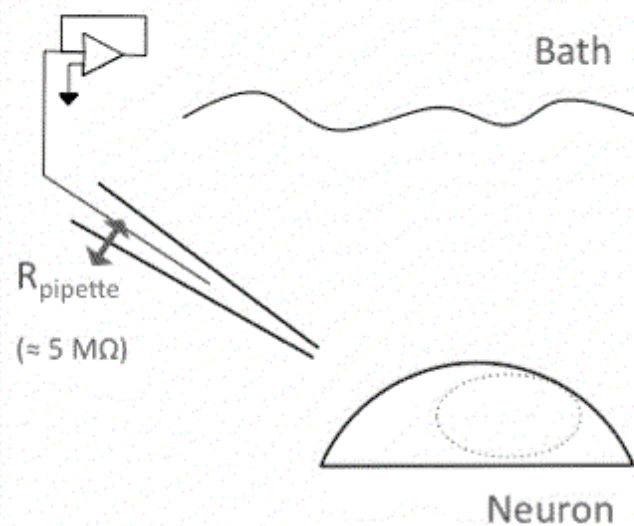


Figure A.31 Patch clamp circuit (see text).

Pipette offset in bath



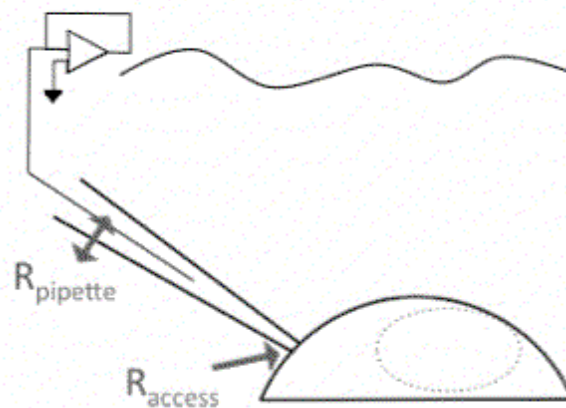
E



I



Cell-attached GΩ seal



$$R_{\text{pipette}} + R_{\text{access}} = R_{\text{series}} (> 1 \text{ G}\Omega)$$



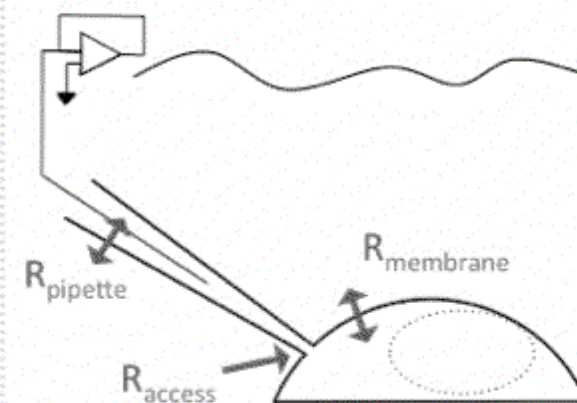
$$I = E / R_s$$

$$\approx 10 \text{ mV} / >1 \text{ G}\Omega$$

$$< 10 \text{ pA}$$



Whole-cell patch clamp

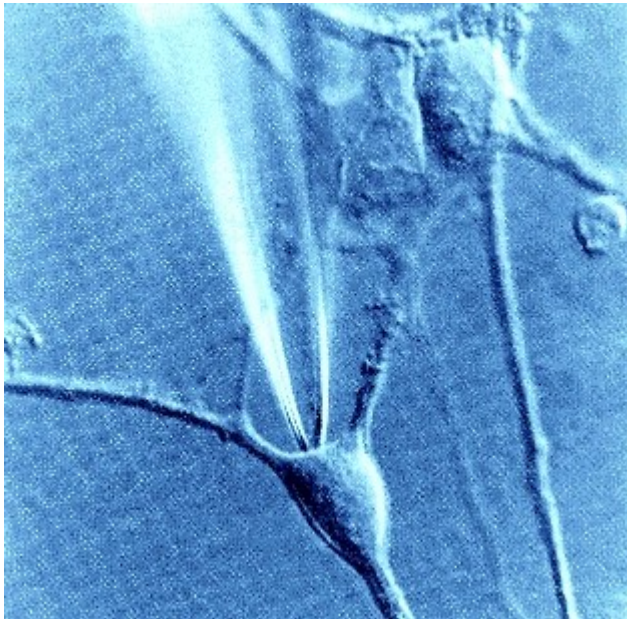


$$R_{\text{pipette}} + R_{\text{access}} = R_{\text{series}} (\approx 15 \text{ M}\Omega)$$

$$R_{\text{series}} + R_{\text{membrane}} = R_{\text{input}} (\approx 800 \text{ M}\Omega)$$



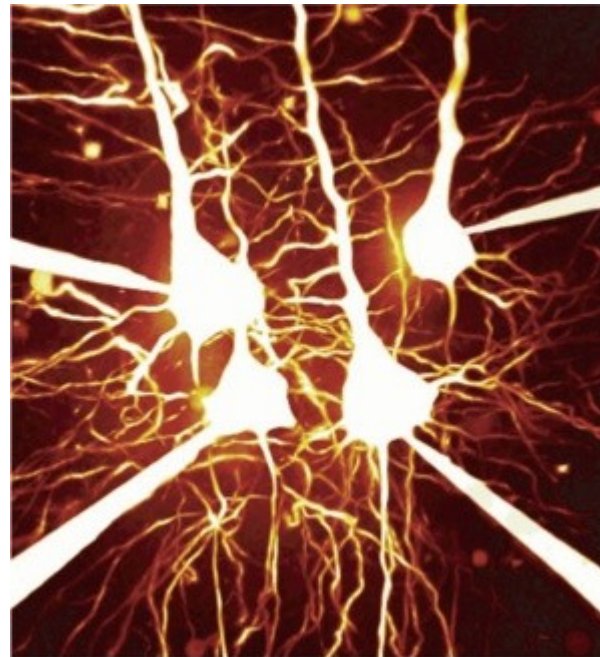
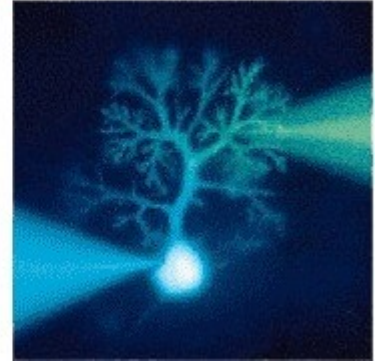
Patch clamp en neuronas

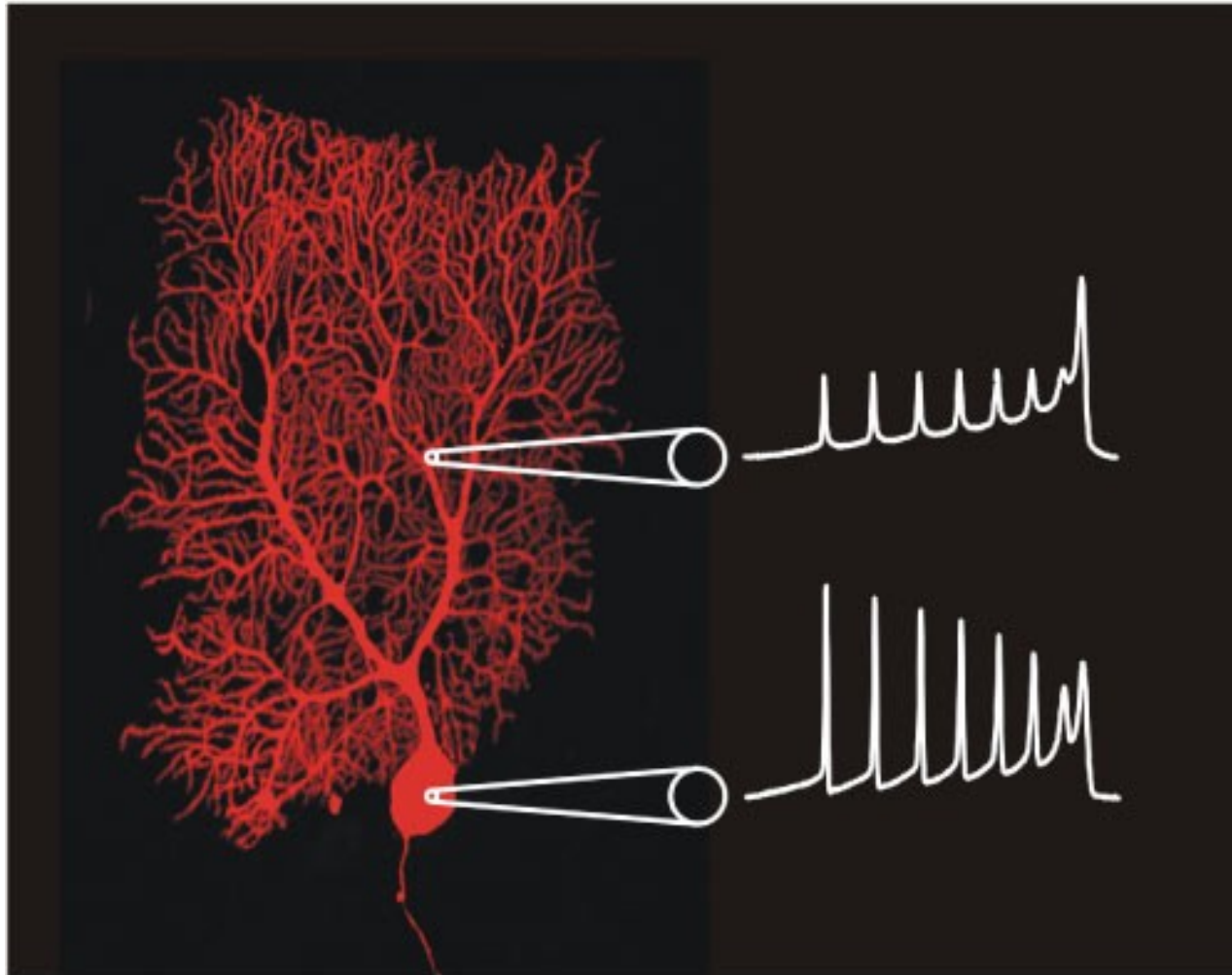


A

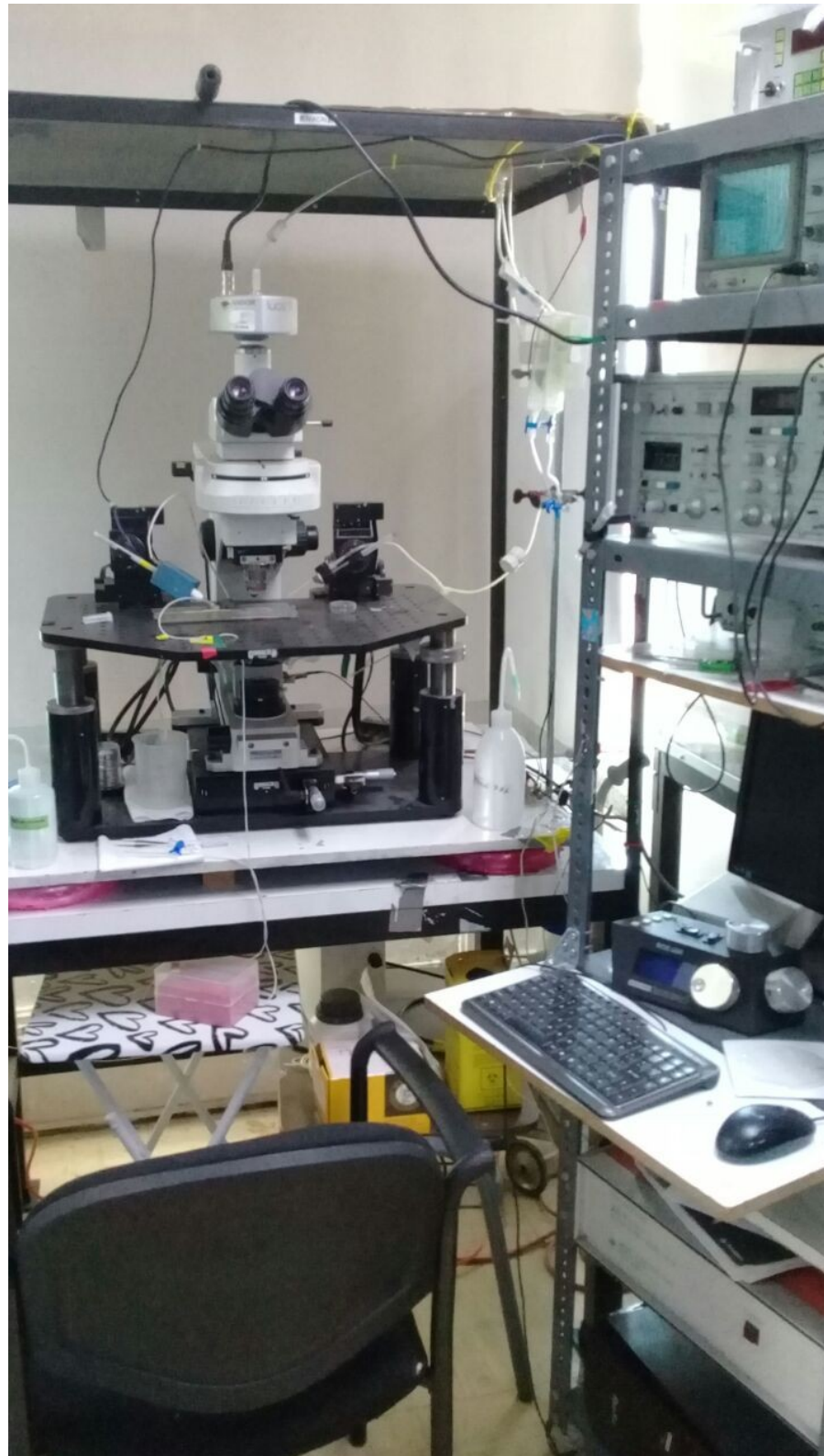


B





Dual soma-dendritic patch recordings from rat cerebellar Purkinje cells are used in conjunction with focal drug ejections to determine the ionic basis of Purkinje cell output.

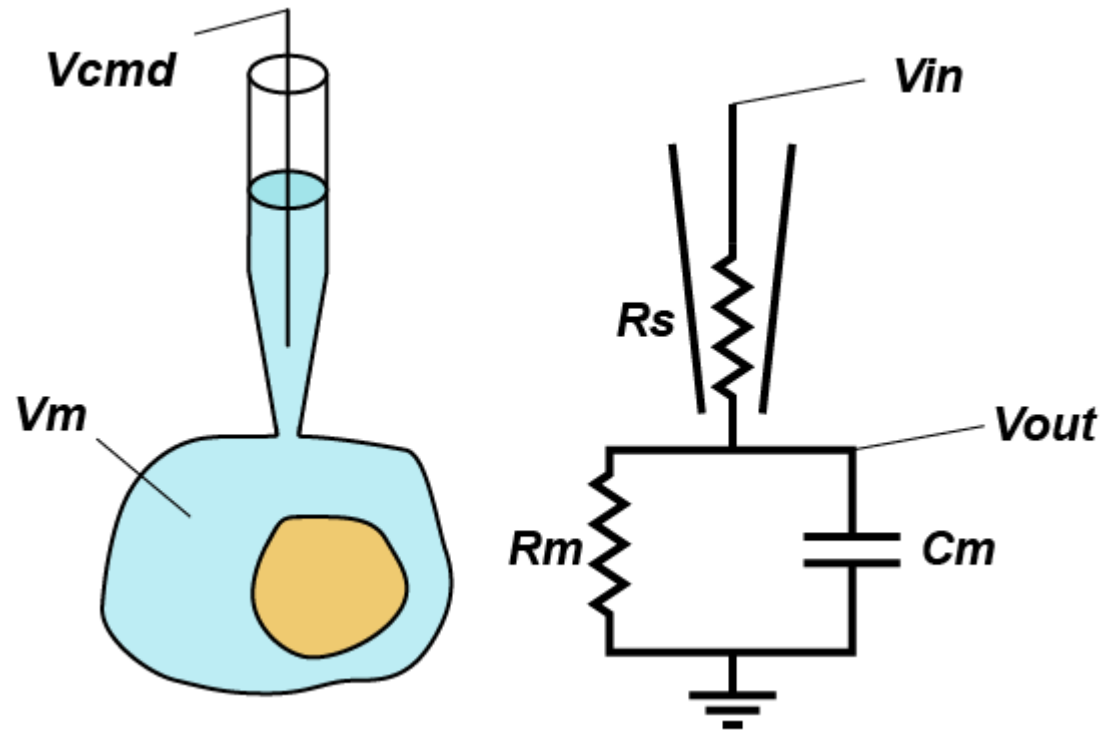


Resistencia en serie



Cell resistance affects patch-clamp experiments...

Resistencia en serie



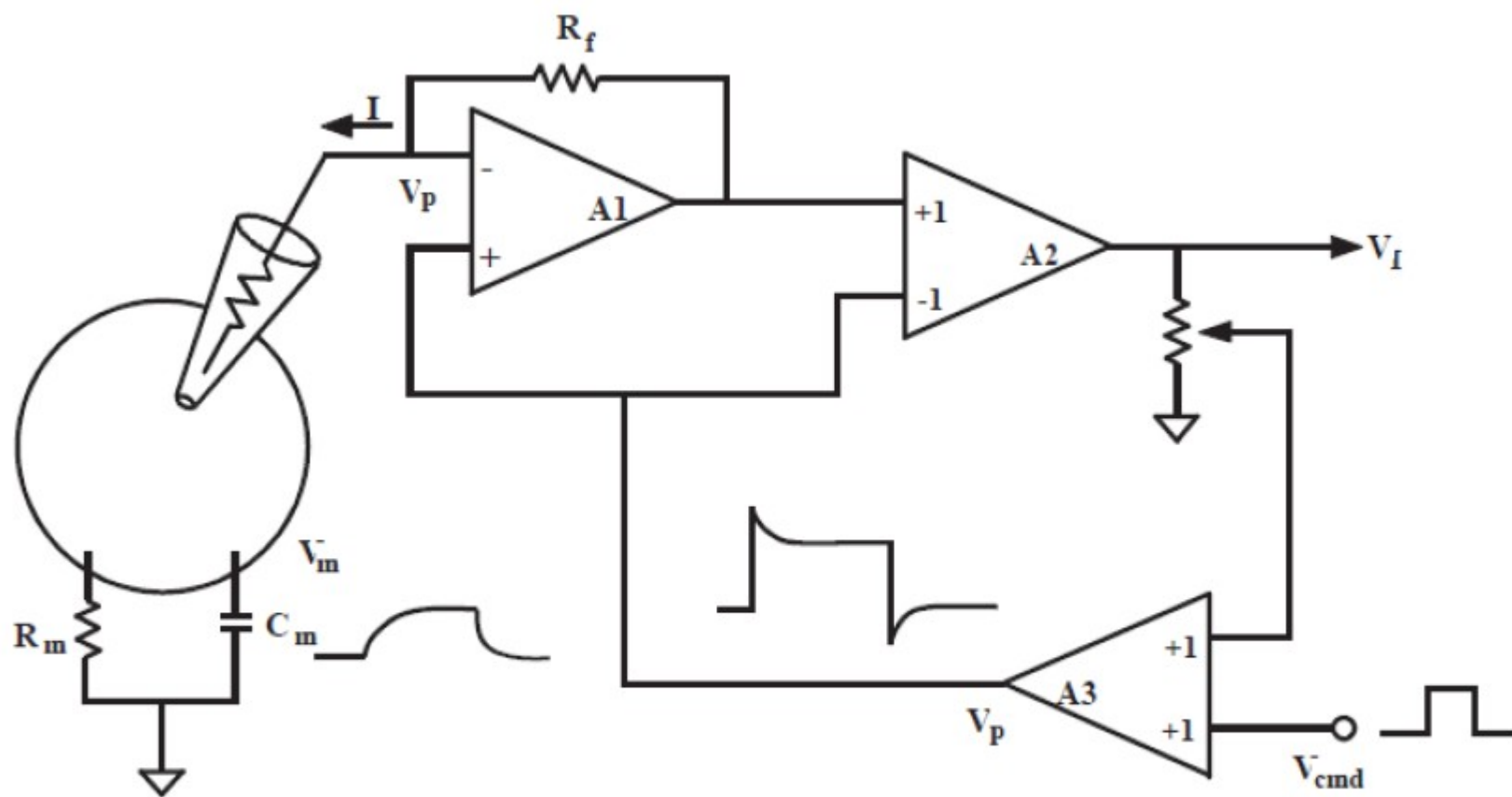


Figure 3-15 Series resistance correction.

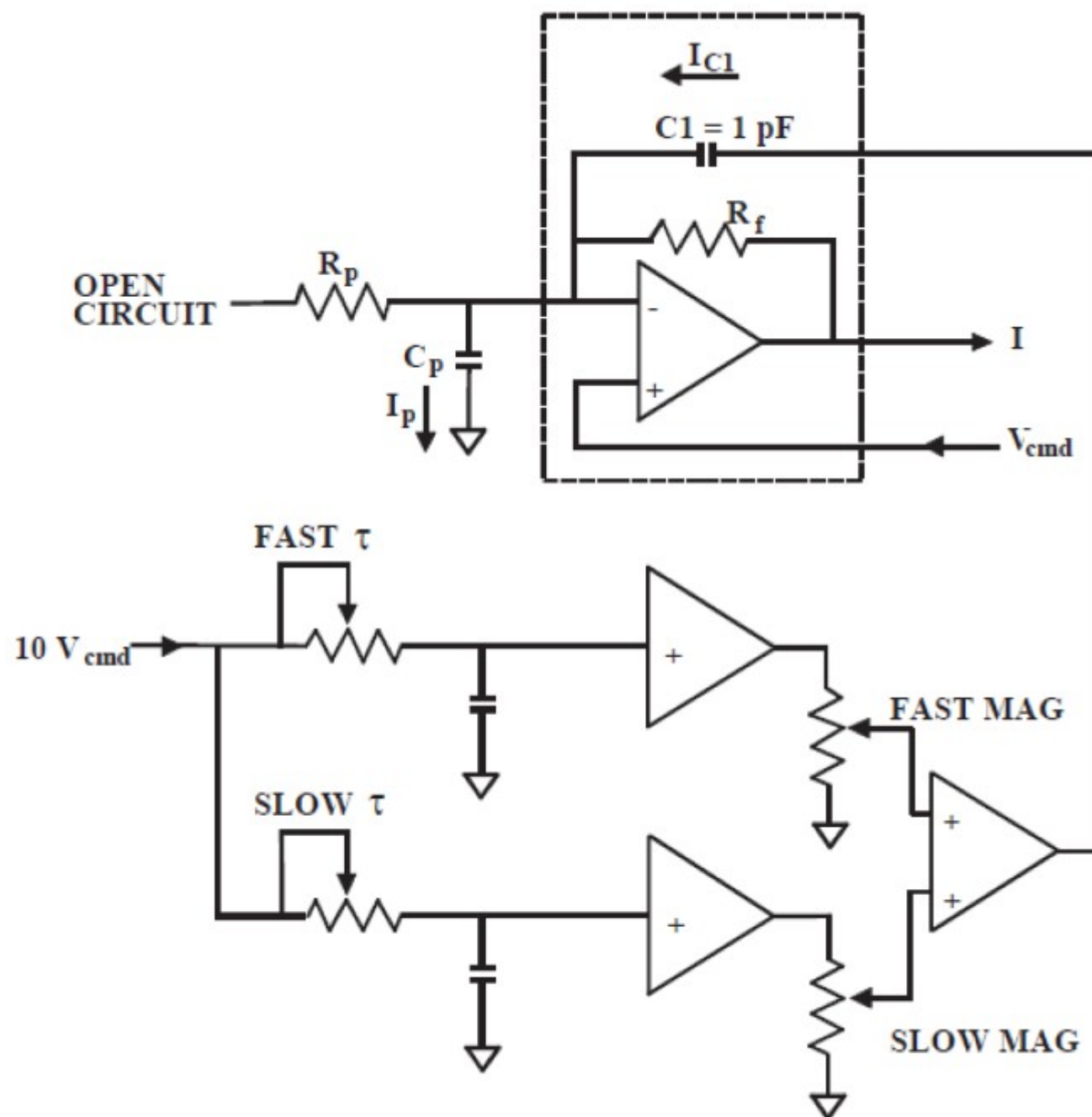


Figure 3-18 Pipette capacitance compensation circuit

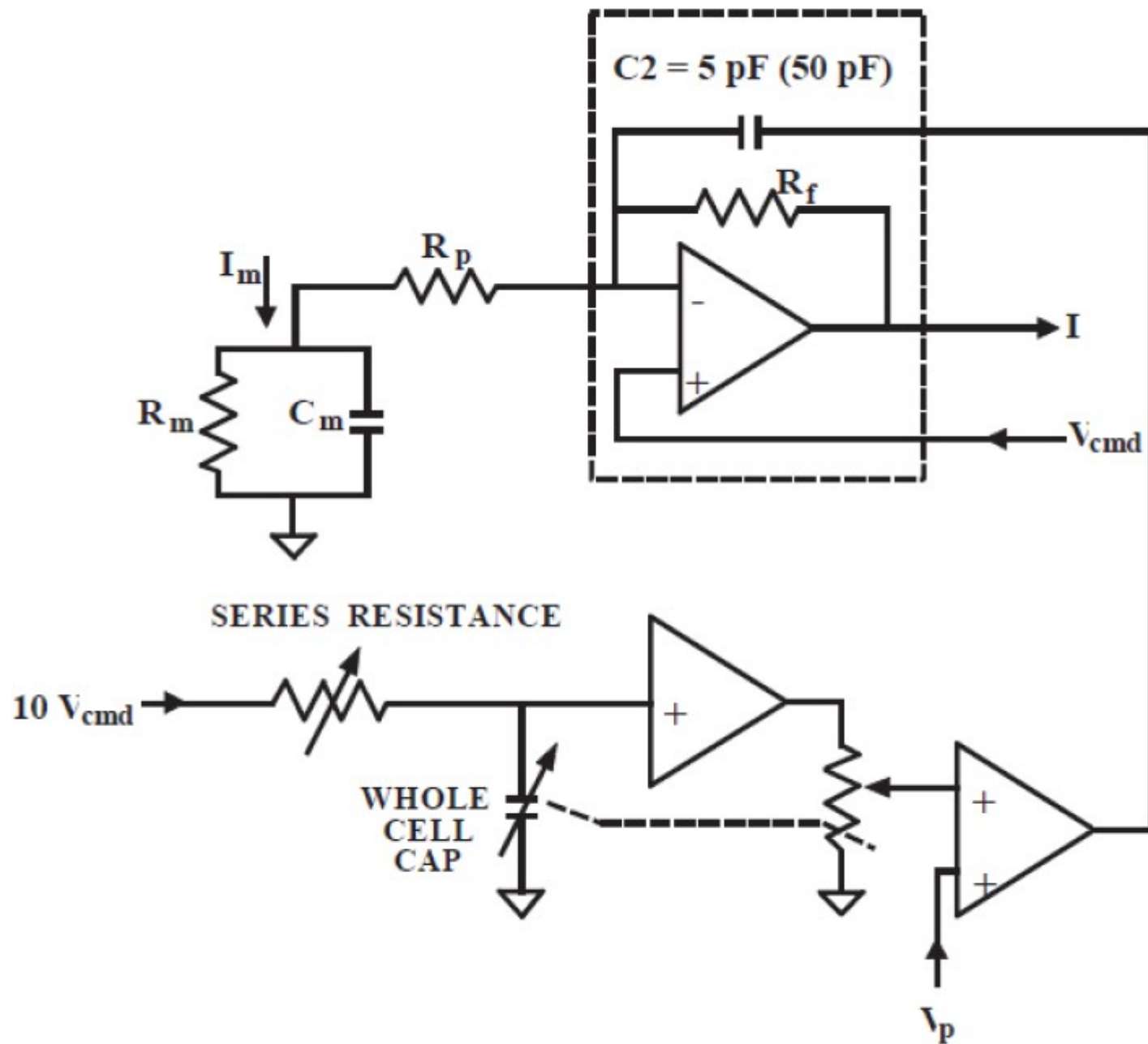
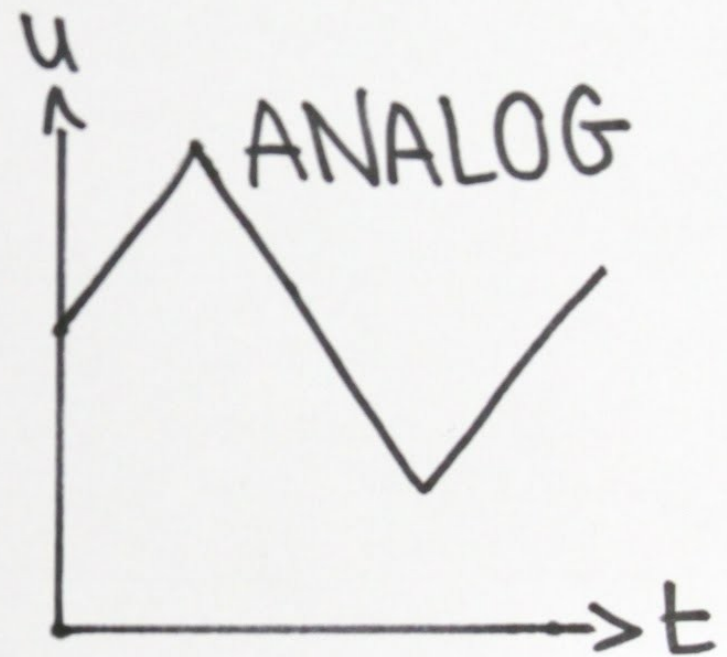
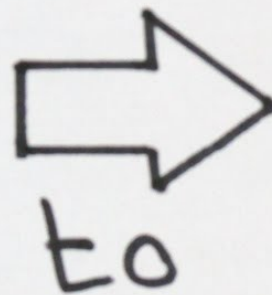
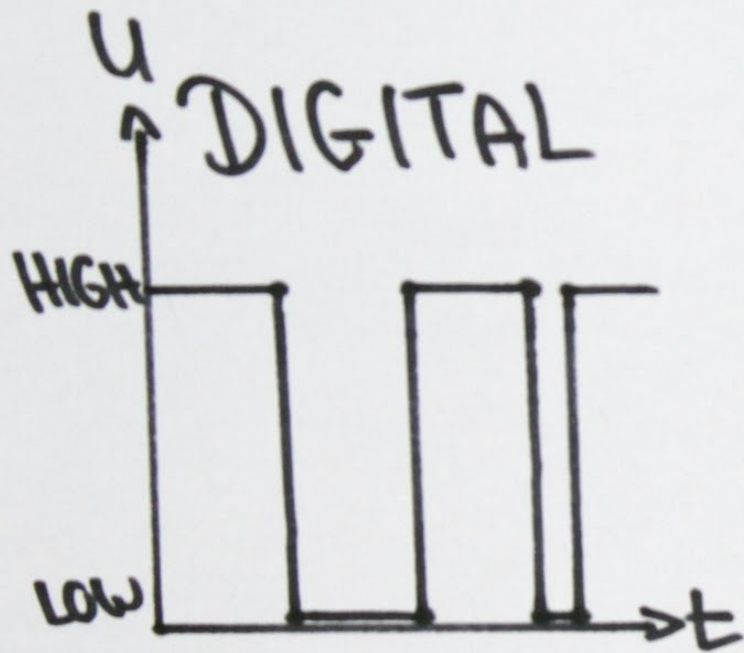


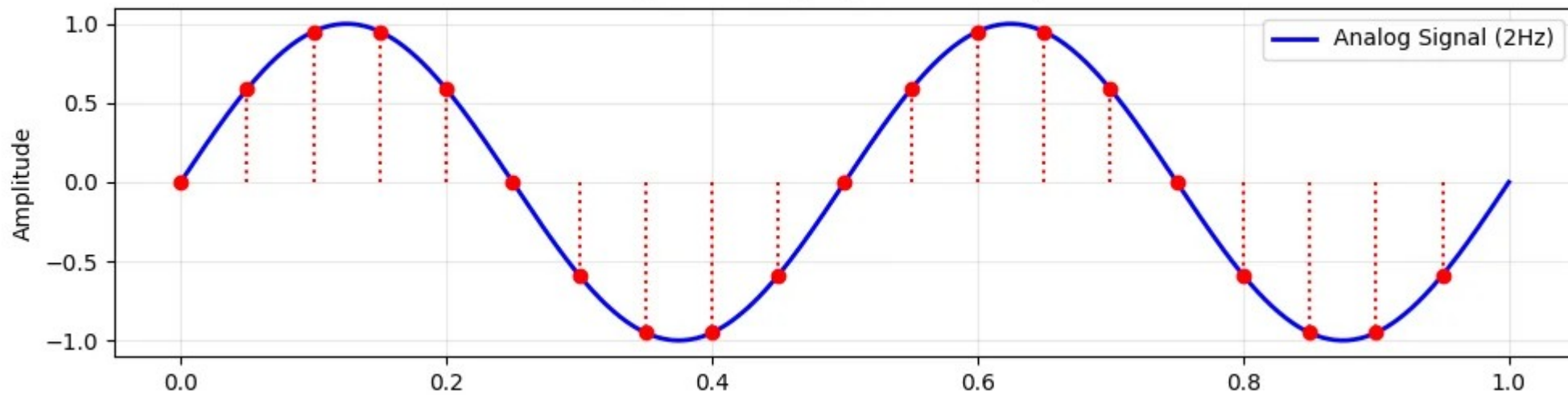
Figure 3-19 Whole-cell capacitance compensation circuit.

Electronic Basics #10:

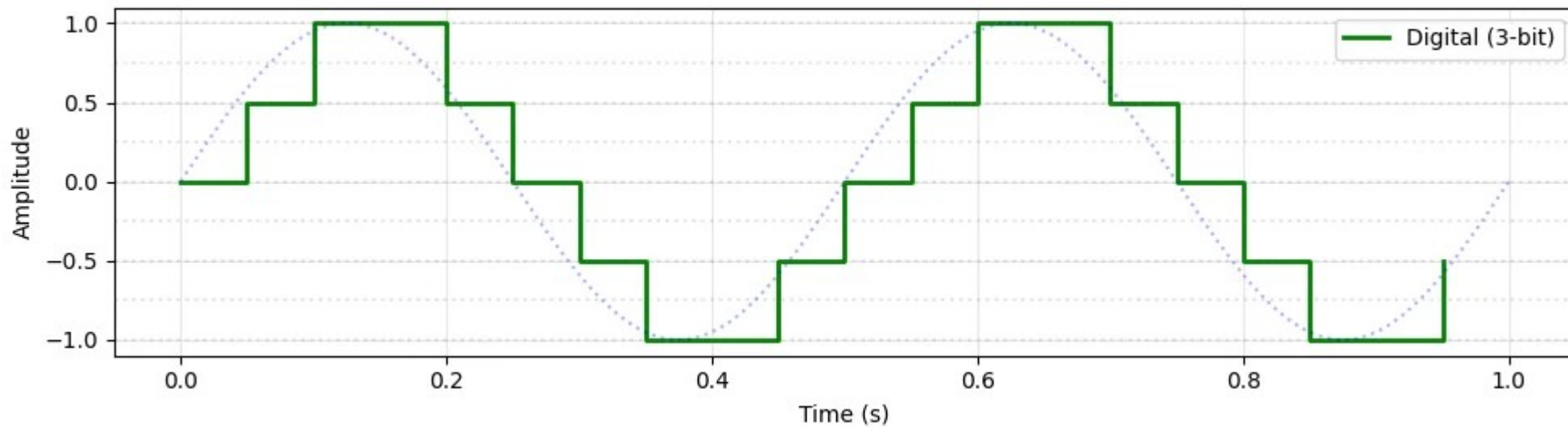
DAC



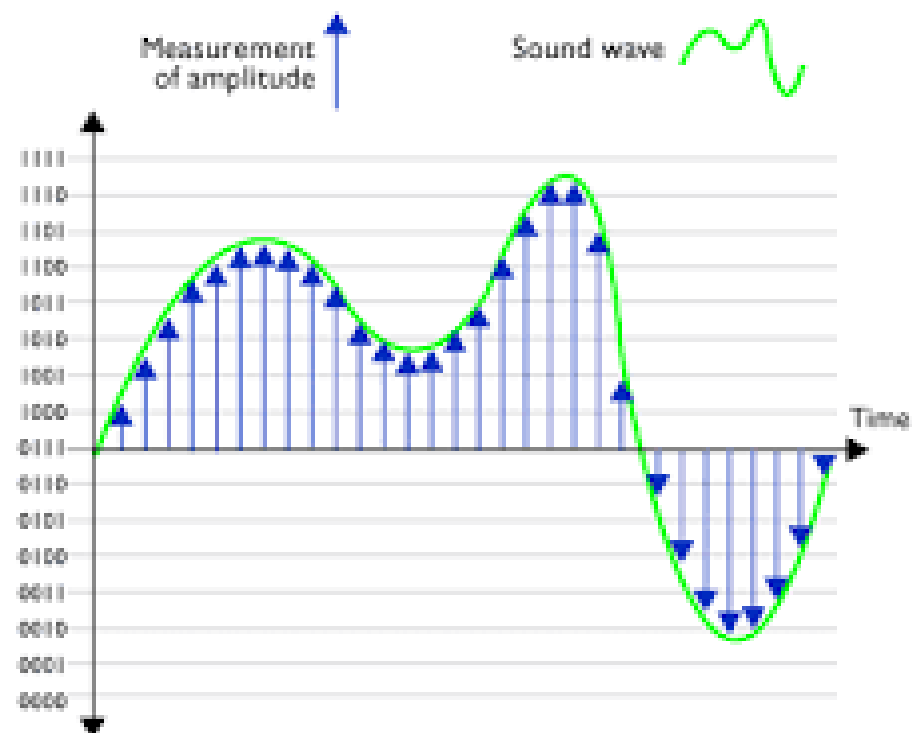
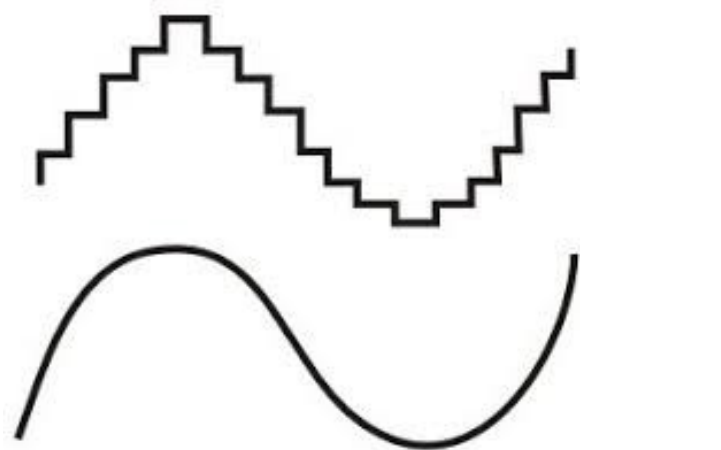
Original Analog Signal with Samples



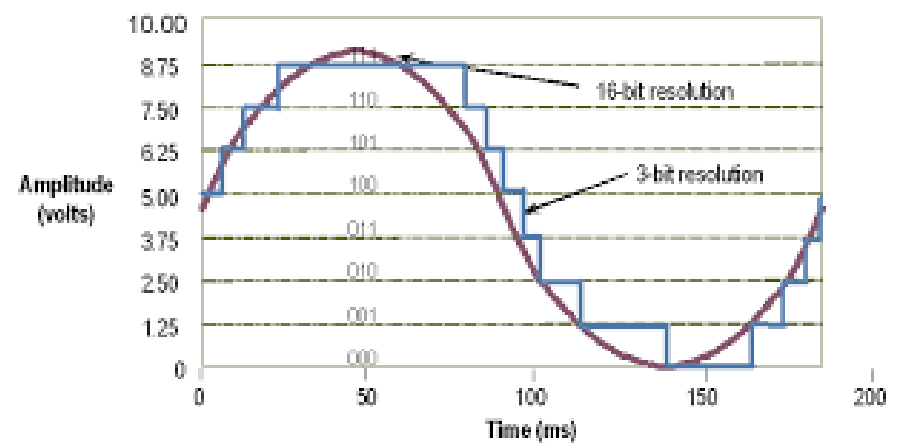
Quantized Digital Output



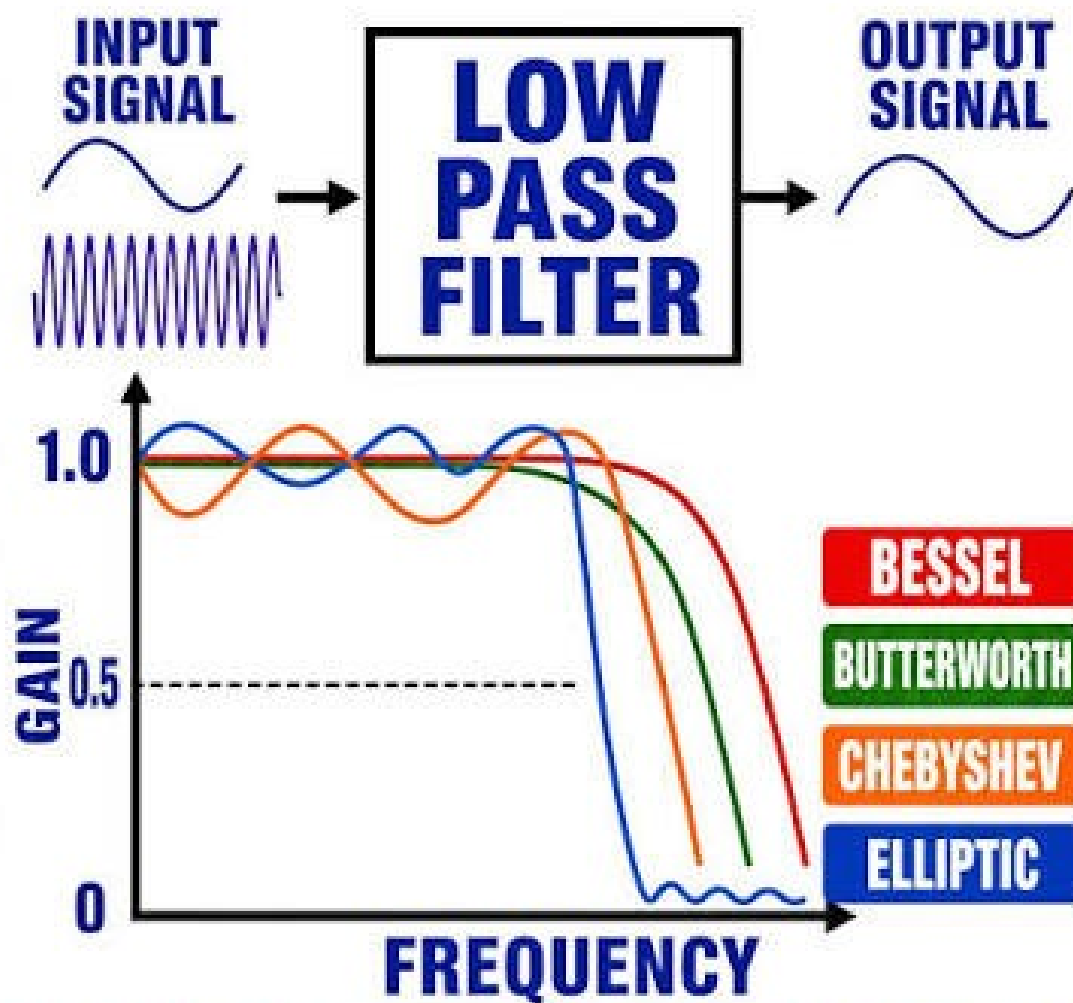
Sampling theorem



Each measurement is assigned a number (byte) according to its amplitude. The end result is a file comprising a string of bytes, eg ...
1001 1110 0001 1010 0111 0100 1111 1101 etc



LOWPASS FILTER TYPES

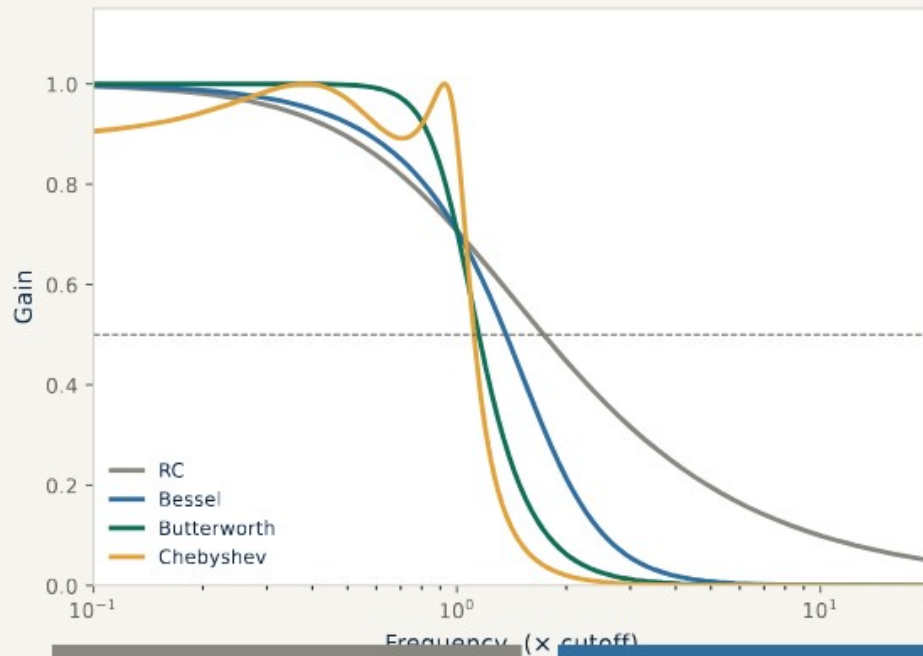


BESSEL	BUTTERWORTH	CHEBYSHEV	ELLIPTIC
USE: Best for signals with linear phase.	USE: General purpose applications.	USE: Sharper cutoff than Butterworth.	USE: Narrowest transition band.

Filter types — frequency response vs. square-pulse response

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Frequency response



RC (1-pole)

Simplest low-pass. Gentle roll-off, used in basic amplifier stages and as the anti-aliasing element.

BESSEL

Best for spike waveforms. Linear phase, no overshoot — preserves timing and shape.

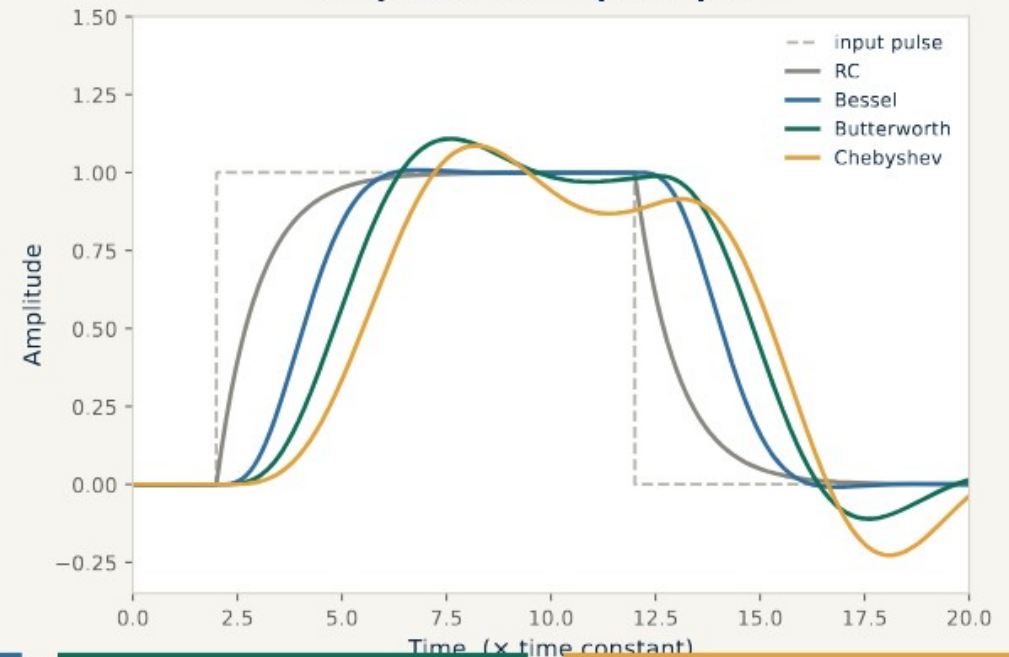
BUTTERWORTH

General purpose. Flat passband, sharper cutoff than Bessel, small overshoot on transients.

CHEBYSHEV

Sharpest cutoff of the three, but pronounced ringing — can distort spike timing.

Response to a square pulse



The sampling theorem — key equations and concepts

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Nyquist-Shannon sampling theorem

A continuous signal can be perfectly reconstructed from its samples only if it contains no frequency components above half the sampling rate.

$$f_s > 2 f_{max}$$

Nyquist criterion for faithful reconstruction

$$f_{Nyq} = \frac{f_s}{2}$$

Nyquist frequency — the reconstruction limit

$$f_{alias} = |f - kf_s|, \quad k = \text{round}(f/f_s)$$

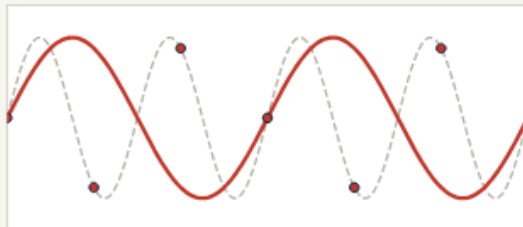
Frequency an under-sampled component is falsely mapped to

Filter <-> acquisition relationship

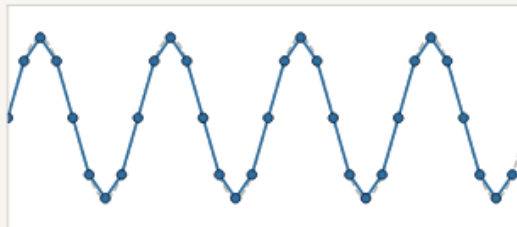
$$f_s \geq 2f_c \Rightarrow f_s \approx (5-10) \times f_c$$

Strict minimum vs. practical rule (real filters roll off gradually)

$f_s < 2 \times f_{max}$ — aliased



$f_s = 8 \times f_{max}$ — faithful



What this means in practice

- Aliasing is a real, false low-frequency signal — indistinguishable from a true one once recorded.
- It must be prevented before digitization; no filter applied afterward can undo it once sampled.
- The anti-aliasing filter sets f_{max} : choose f_c for the biology of interest, then derive f_s from it.
- Bare Nyquist ($f_s = 2 \times f_c$) is a mathematical minimum, not safe practice — real filters roll off gradually, hence the 5-10x margin.

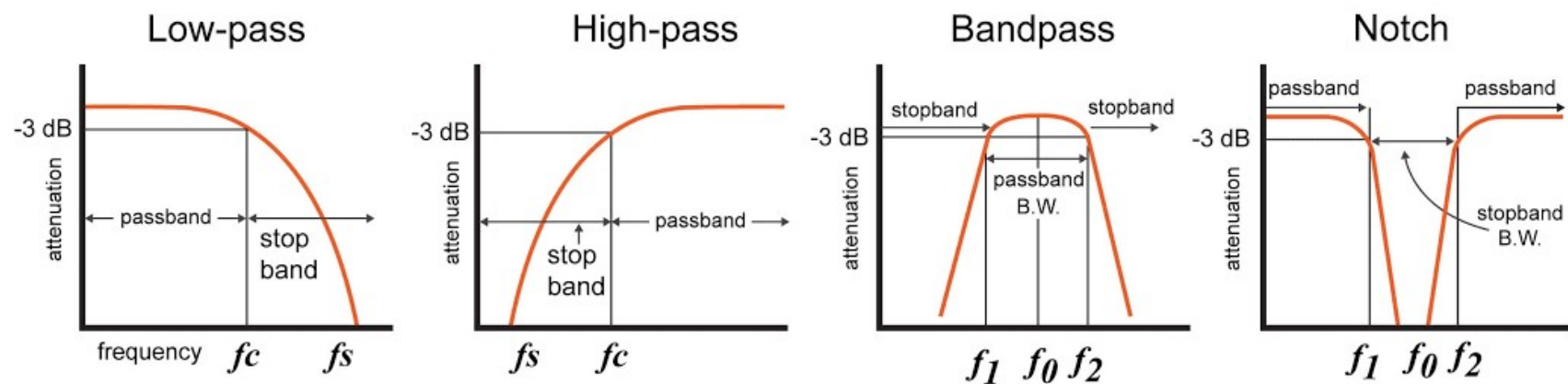
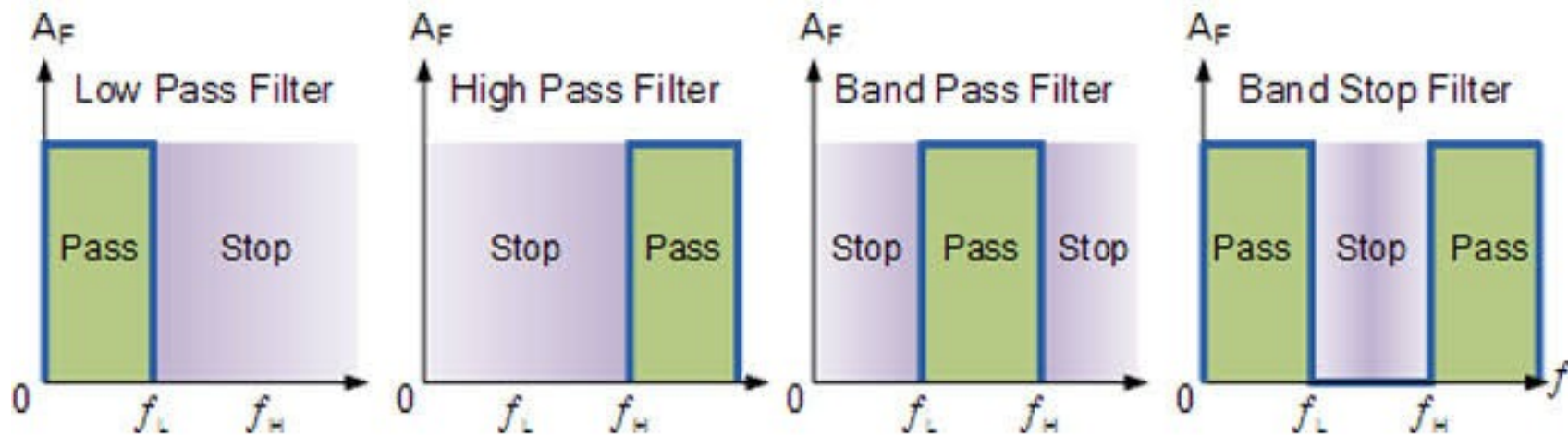
Quick reference

Signal

LFP / field potentials
Synaptic currents (VC)
Extracellular spikes

f_{max} needed

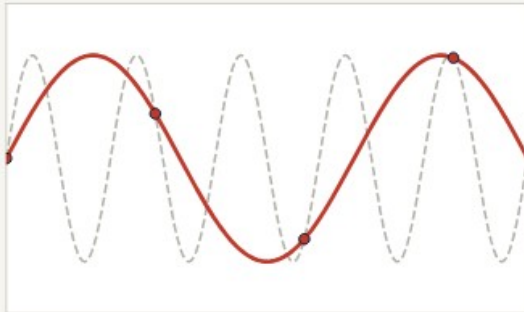
~300 Hz
2-5 kHz
3-10 kHz



Sampling simulations — effect of f_s/f_{max} on signal reconstruction

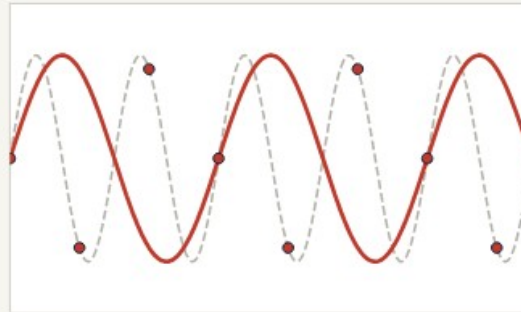
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$f_s = 0.7 \times f_{max}$ — SEVERE ALIASING



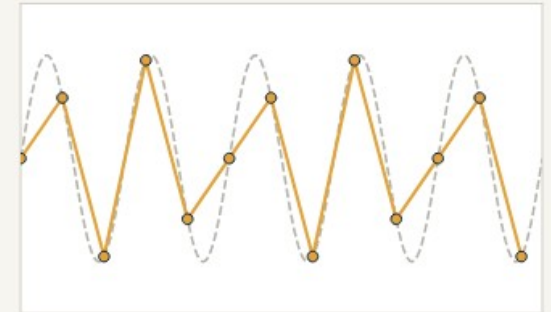
4 samples across 5 cycles

$f_s = 1.5 \times f_{max}$ — ALIASED



8 samples across 5 cycles

$f_s = 2.5 \times f_{max}$ — MARGINAL



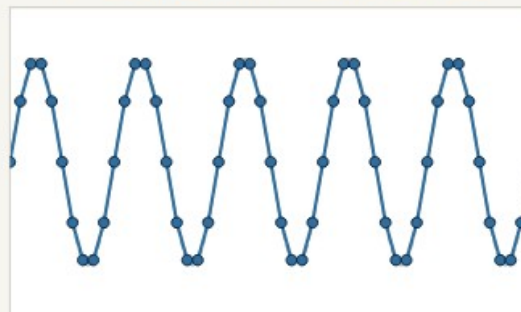
13 samples across 5 cycles

$f_s = 5 \times f_{max}$ — GOOD (rule-of-thumb)



25 samples across 5 cycles

$f_s = 10 \times f_{max}$ — EXCELLENT



50 samples across 5 cycles

Takeaway

- $f_s < 2 \times f_{max}$**
false, slow signal (alias)
- $f_s = 2-2.5 \times f_{max}$**
borderline, distorted
- $f_s = 5-10 \times f_{max}$**
faithful reconstruction

The anti-aliasing filter sets $f_{max} = f_c$ before the ADC ever sees the signal.